

Physical Chemistry (I)

Lecture 16

2차 중간고사 안내 및 복습



2차 중간고사 안내

- 시간: 5월 13일(화) 오전 10:30~11:30
- 장소: 화학관 112호(홀수 학번), 315호(짝수 학번)
- 신분증 지참(모바일 학생증도 가능)
- 계산기 불필요

시험 범위

- 5장 "헬름홀츠 및 기브스 에너지"
- 6장 "통계역학"
- 7장 "상 전이"
- 8장 "단순 혼합물" 중 진도 나간 곳까지

시험 문제

총 여섯 문제

- 과제 문제 혹은 그 변형(3문제)
 - 유도라면 중간 단계를 주지 않을 수도 있음
- 수업 시간에 배운 식 유도 및 응용(3문제)
 - 강의 자료 기준

Chapter 5

Thermodynamic potentials

Clausius ineq.

First law

Internal energy

$$U$$

Helmholtz energy

$$A = U - TS$$

Enthalpy

$$H = U + pV$$

Gibbs energy

$$G = H - TS$$

Spontaneity

$$\Delta A_{T,V} \leq 0$$

$$\Delta G_{T,p} \leq 0$$

Maximum work

$$A_T = w_{\max}$$

$$G_{T,p} = w_{\text{add,max}}$$

Differential forms

Applications

- Gibbs-Helmholtz equation
- Maxwell relations
- Thermodynamic equation of state

Thermochemistry

Pressure dependence

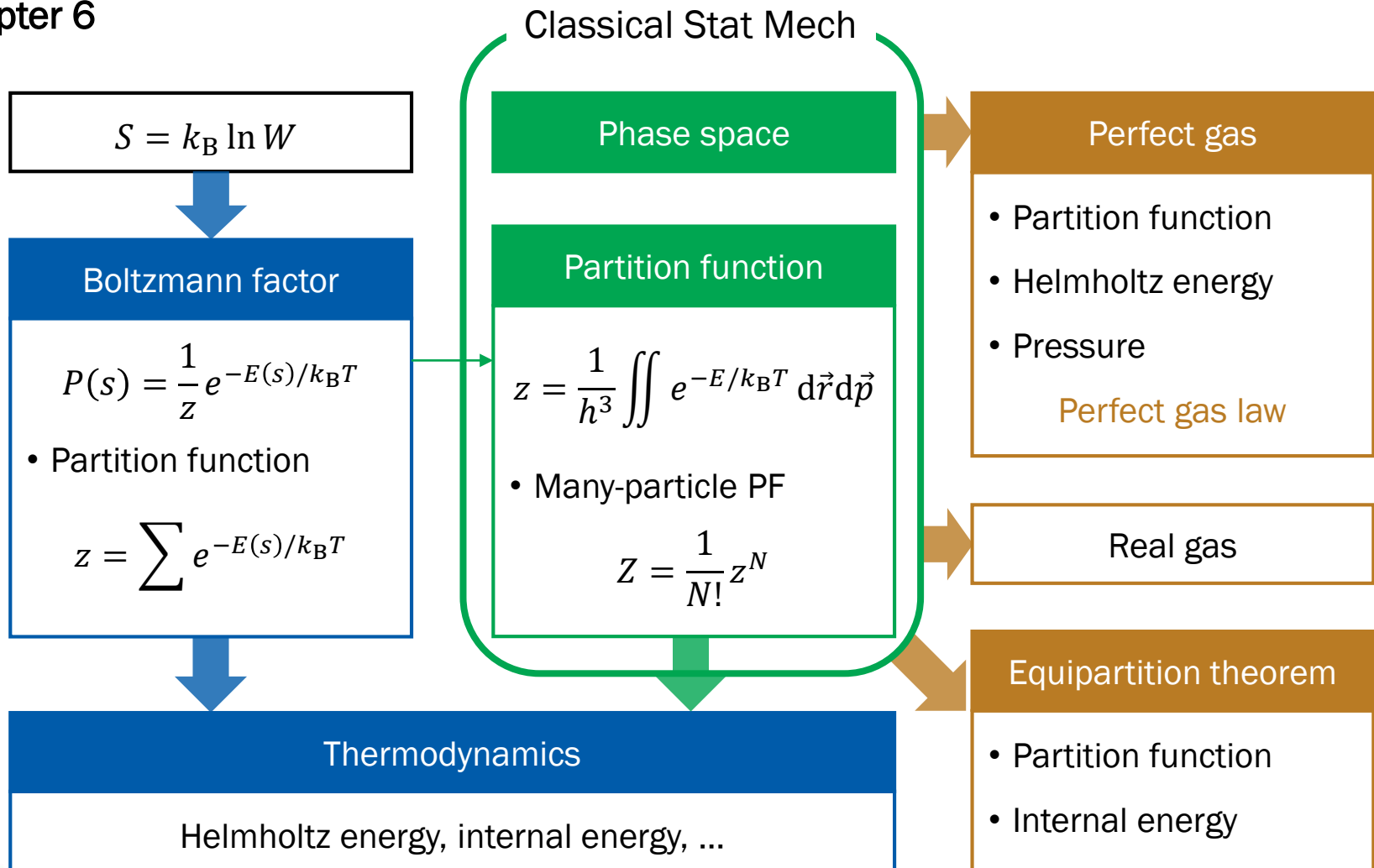


Concepts and Derivations in Chpt 5

- [C] Helmholtz and Gibbs energies
- [D] Spontaneity: Clausius inequality
- [D] Meaning of Helmholtz and Gibbs energies
- [D] Thermodynamic potentials
- [D] Gibbs-Helmholtz equation
- [D] Maxwell relations
- [D] Thermodynamic equation of state
- [D] Pressure dependence of Gibbs energy



Chapter 6



Concepts and Derivations in Chpt 6

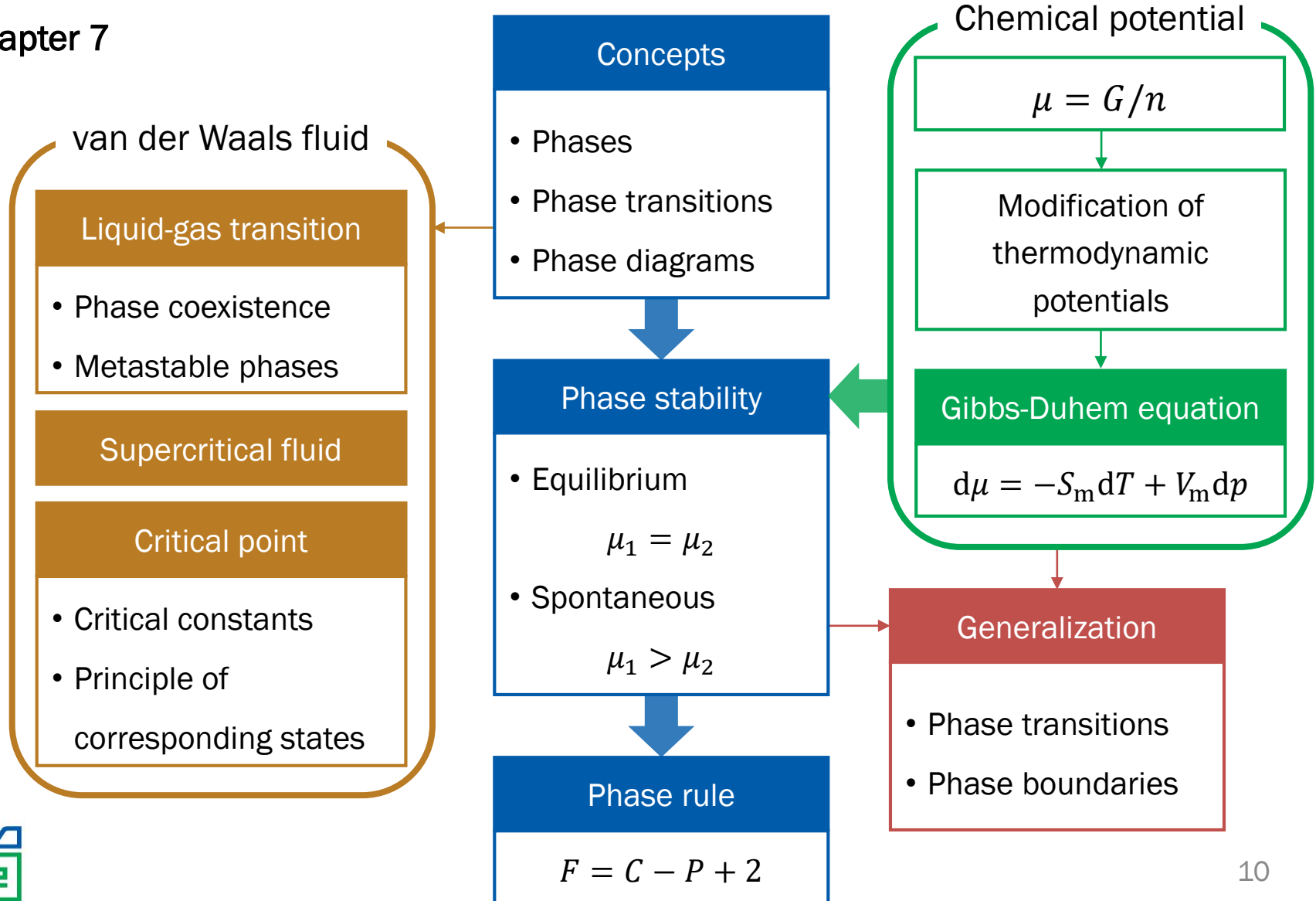
- [D] Boltzmann factor
- [C] Partition function
- [D] Partition function and thermodynamics
 - [D] Internal energy
 - [D] Shannon entropy
 - [D] Helmholtz energy
 - [D] Other thermodynamic quantities
 - [D] Example: spin system
- [C] Phase space



Concepts and Derivations in Chpt 6

- [C] Partition function of classical system
 - [D] 1-dimensional spring
 - [D] N -particle partition function
 - [C] Indistinguishability
- [D] Perfect gas: perfect gas law
- [C] Real gas: van der Waals equation
- [D] Equipartition theorem

Chapter 7



Concepts and Derivations in Chpt 7

- [C] Phases and phase diagrams
- [C] Phase transitions
- [C, D] Chemical potential
- [D] Rewriting thermodynamic potentials
- [D] Gibbs-Duhem equation
- [C, D] Phase rule
- [C] Liquid-gas transitions



Concepts and Derivations in Chpt 7

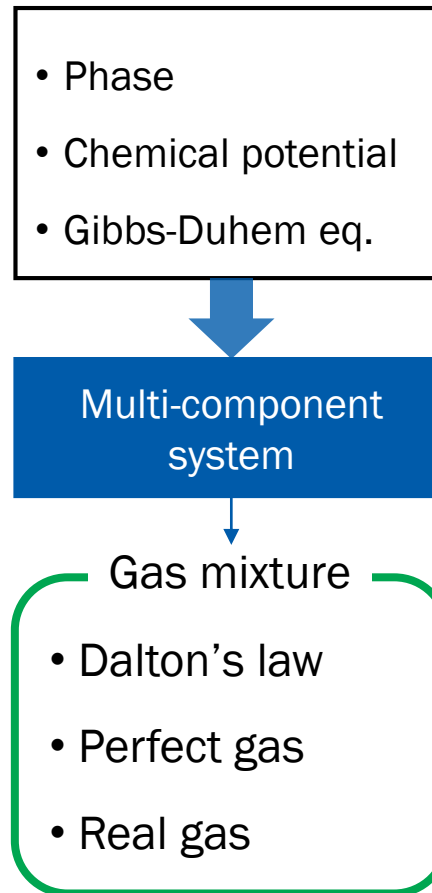
- [C] vdW: Liquid-gas transitions
 - [C] Maxwell construction
 - [D] Lever rule
 - [C] Binodal and spinodal lines
- [C] vdW: Supercritical fluid
- [C] vdW: Critical point
 - [D] Critical constants
 - [D] Critical compression factor
 - [D] Reduced variables
 - [C] Principle of corresponding states



Concepts and Derivations in Chpt 7

- [D] T - and p -dependences of chemical potentials
- [C] Phase boundaries
 - [D] Clausius equation
 - [D] Clausius-Clapeyron equation
 - [C] Interpretations

Chapter 8



Concepts and Derivations in Chpt 8

- Extension to multi-component systems
 - [C] Thermodynamic potentials
 - [C, D] Partial molar quantities
 - [C, D] Chemical potential
 - [D] Gibbs-Duhem equation
- [C] Dalton's law
- [D] Mixing of perfect gases
- [C] Real gases: fugacity and fugacity coefficient
- [D] Mixing of real gases

